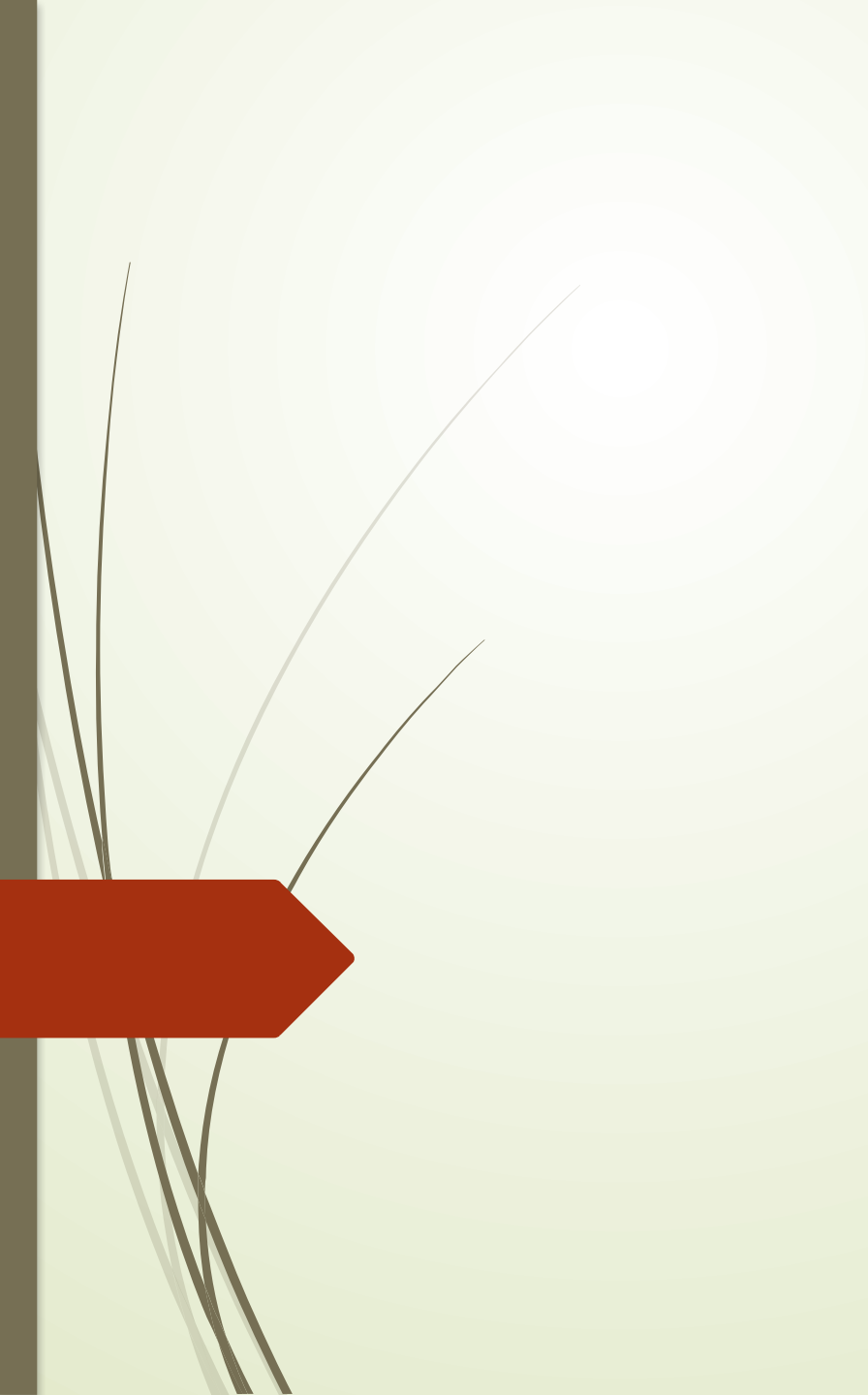




PHP Practicum

IS4102

Presentation 01

WGS Lakshitha
Instructor in Computer Technology
Faculty of Computing

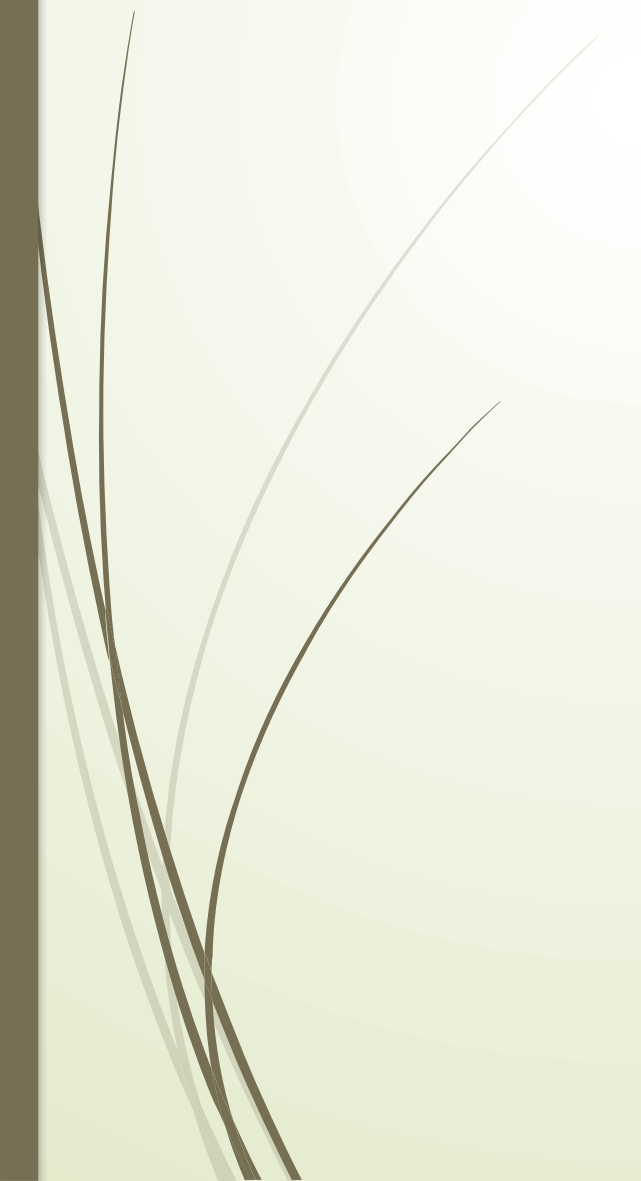


About PHP

- ❑ PHP is an interpreted language.
 - ❑ You write a series of statements.
 - ❑ Apache hands these statements to the PHP interpreter.
 - ❑ The interpreter executes these statements one by one.
 - ❑ When it find an error, it stops running and signals the error.
 - ❑ PHP 5.3 is the current version.
 - ❑ Compiled languages are different. They read the whole program before starting to execute it.
- 



Ingredients



Web Server

Database

PHP Parser

Tool Packages for Developing PHP Pages

- WAMP



www.wampserver.com

- XAMPP



XAMPP

Basic PHP Syntax

A PHP scripting block always starts with `<?php` and ends with `?>`

```
<html>
```

```
<body>
```

```
<?php
```

```
    echo "My first PHP script!";
```

```
?>
```

```
</body>
```

```
</html>
```

Output: My first PHP script!

How to save a PHP file?

hello.php

File name _____

File extension/Format _____

Comments in PHP

A comment in PHP code is a line that is not executed as a part of the program. Its only purpose is to be read by someone who is looking at the code.

1) Single Line Comment

```
<?php  
    //This is a single line comment  
?>
```

```
<?php  
    #This is a one-line shell-style comment  
?>
```

Comments in PHP(Con...)

2) Multi Line Comment

```
<?php  
  
    /* This is a multi-line comment  
    yet another line of comment */  
  
?>
```


Variables in PHP

All variables in PHP start with a **\$ sign symbol**. It's may contain strings, numbers, or arrays.

`name = "John";` ← **incorrect**
`$name = "John";` ← **correct**

`X = 5;` ← **incorrect**
`$X = 5;` ← **correct**

`Var = 5;` ← **incorrect**
`$Var = 5;` ← **correct**
`$_Var = 5;` ← **correct**

PHP variable name shouldn't contain spaces. If a variable name should be more than one word, it should be separated with an underscore.

`$First Name = "John";` ← **incorrect**
`$First_Name = "John";` ← **correct**

Variable Examples

```
<?php
    $x = 5; //numeric values
    $y = 8; //numeric values
?>
```

← Numeric variables

```
<?php
    $name = "John"; //string values
    $city = "Colombo"; //string values
?>
```

← String variables



Exercise 01

- 1) Define variables 'first_name' and 'last_name' using PHP. Assign your name to these variables, then print your full name?
 - 2) Write PHP code to add two integer values?
- 

Answer 01

```
<?php
// Define the two integer values
$first_name = "kasun";
$last_name = "Perera";

// Print full name
echo "My full name is: $first_name $last_name";
?>
```

Answer 01(Con...)

```
<?php
// Define the two integer values
$number1 = 10;
$number2 = 5;

// Add the two numbers
$sum = $number1 + $number2;

// Output the result
echo "The sum of $number1 and $number2 is: $sum";
?>
```

Loops in PHP

- A loop in PHP is used to execute a statement or a block of statements multiple times until a specific condition is met.
- PHP supports four types of looping,
 - for loop
 - while loop
 - do-while loop
 - foreach loop



for loop

Executes a block of code a specified number of times.

```
for (initialization expression; test condition; update expression) {  
    // code to be executed  
}
```

Exercise 02

- Write PHP code to print the first five numbers?

```
<?php
    for ($i = 0; $i < 5; $i++) {
        echo "Iteration: $i <br>";
    }
?>
```


Exercise 03

- Write PHP program to calculate the sum of numbers from 1 to 50?

```
<?php
$sum = 0;
for ($i = 1; $i <= 50; $i++) {
    $sum += $i;
}
echo "The sum of numbers from 1 to 100 is:
$sum";
?>
```

Exercise 04

- Write PHP program to print even numbers from 2 to 20?

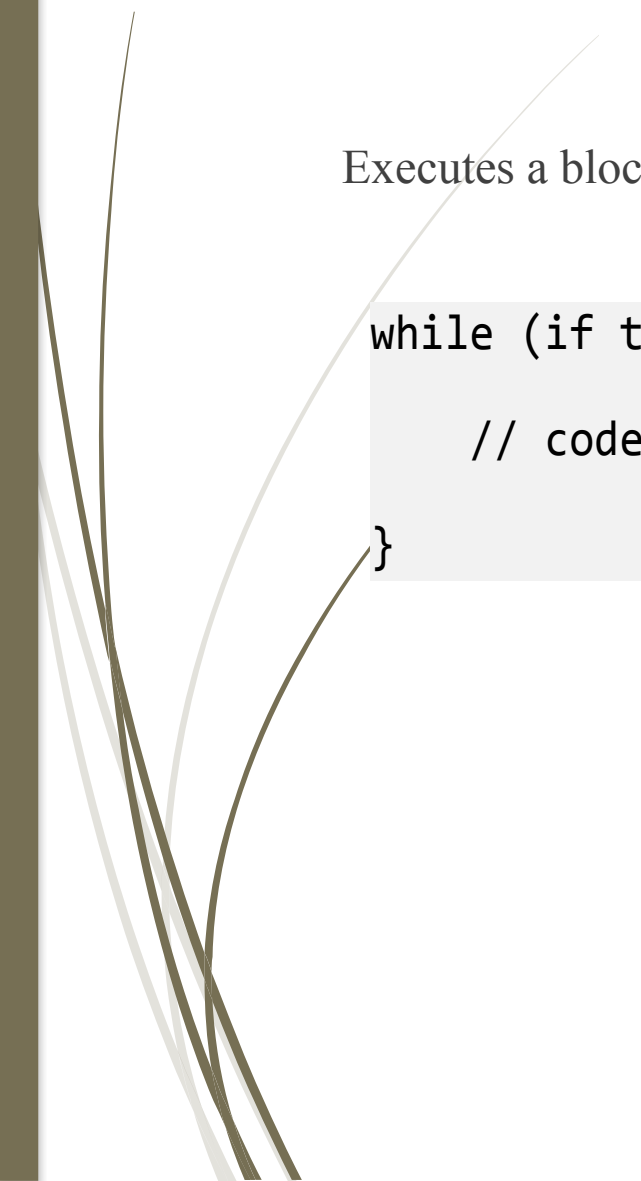
```
<?php
for ($i = 2; $i <= 20; $i += 2) {
    echo $i . "<br>";
}
?>
```



while loop

Executes a block of code as long as the specified condition is true.

```
while (if the condition is true) {  
    // code is executed  
}
```



Exercise 05

- Write PHP program to print numbers from 1 to 10? (note: use only while loop)

```
<?php
$i = 1;
while ($i <= 10) {
    echo $i . "<br>";
    $i++;
}
?>
```

Exercise 06

- Write PHP program to print even numbers from 2 to 20 (note: use only while loop)

```
<?php
$i = 2;
while ($i <= 20) {
    echo $i . "<br>";
    $i += 2;
}
?>
```



do-while loop

Executes a block of code once, then repeats the loop as long as the specified condition is true.

```
do {  
    //code is executed  
} while (if condition is true);
```

Exercise 07

- Write PHP program to print numbers from 1 to 10 (note: use only do-while loop)

```
<?php
$i = 1;
do {
    echo $i . "<br>";
    $i++;
} while ($i <= 10);
?>
```

Exercise 08

- Write PHP program to print numbers in reverse order from 10 to 1 (note: use only do-while loop)

```
<?php
$i = 10;
do {
    echo $i . "<br>";
    $i--;
} while ($i >= 1);
?>
```


foreach loop

Iterates over arrays and objects.

Iterating over an array and printing each element,

```
<?php  
$colors = array("red", "green", "blue");  
foreach ($colors as $color) {  
    echo $color . "<br>";  
}  
?>
```



Exercises



- ❑ Write a PHP script that prints the multiplication table of a given number (e.g., 5) up to 10?
- ❑ Write a PHP script to calculate the factorial of a given number using a for loop?
- ❑ Write a PHP script to print the first 10 terms of the Fibonacci sequence using a for loop?
- ❑ Write a PHP script that takes a number and finds the sum of its digits using a while loop?



Thank you!